🔷 Enterprise AI & Data Engineering Quiz (Advanced – B.Tech Level)

Section 1: Azure ML & AIOps

Q1.

You are building a model in Azure Machine Learning and want to ensure reproducibility across environments. What should you use?

A. Manual scripts
B. Environment configuration + versioned datasets
C. Local machine setup
D. Random configurations

Q2.

A model needs to retrain automatically when new data arrives daily. What is the best solution?

A. Manual retraining
B. Azure ML pipelines with scheduled triggers
C. Ignore new data
D. Delete old model

Q3.

An IT operations team wants to detect anomalies in server logs using AI. This is an example of:

A. DevOps
B. AIOps
C. Data warehousing
D. ETL

Q4.

Which component helps track experiments, metrics, and model versions?

A. Azure Storage
B. MLflow / Experiment tracking
C. Virtual Machines
D. Power BI

Section 2: Azure AI Foundry & Enterprise Deployment

Q5.

A company wants to deploy multiple AI applications with centralized governance using Azure AI Foundry. What is the best approach?

A. Independent deployments
B. Centralized AI platform with shared services
C. Manual deployments
D. Local hosting

Q6.

You need to integrate AI models with an existing ERP system. What is the best practice?

A. Replace ERP
B. API-based integration layer
C. Direct DB access
D. Manual data transfer

Q7.

A chatbot built using Azure AI Foundry must fetch real-time order data. Which integration method is best?

A. Static responses
B. REST API integration
C. CSV uploads
D. Manual lookup

Q8.

Which deployment pattern is best for enterprise AI scalability?

A. Monolithic deployment
B. Microservices-based AI architecture
C. Local scripts
D. Offline processing

Section 3: Enterprise Integration & Best Practices

Q9.

An enterprise AI system must scale during peak usage. What should be implemented?

A. Fixed resources
B. Auto-scaling and load balancing
C. Manual scaling
D. No scaling

Q10.

To ensure compliance and governance in AI systems, what is essential?

A. No logging
B. Audit trails + RBAC
C. Open access
D. No documentation

Q11.

Your AI system is deployed globally but facing latency issues. What is the best solution?

A. Single region deployment
B. Multi-region deployment
C. Reduce users
D. Stop system

Q12.

To reuse AI components across projects, what should be done?

A. Build from scratch
B. Create modular reusable services
C. Avoid reuse
D. Hardcode logic

Section 4: Azure Data Factory (ADF Fundamentals)

Q13.

In Azure Data Factory, what is a pipeline?

A. Storage unit
B. Group of activities for data movement/transformation
C. Database
D. Dashboard

Q14.

Which component defines connection to data sources?

A. Dataset
B. Linked Service
C. Pipeline
D. Trigger

Q15.

A dataset in ADF represents:

A. Compute resource
B. Structure of data
C. Pipeline logic
D. Trigger

Q16.

You want to copy data from SQL to Blob storage. Which activity is used?

A. Lookup
B. Copy Activity
C. Delete Activity
D. Web Activity

Section 5: Pipelines, Triggers & Integration Runtime

Q17.

A pipeline should run every day at midnight. Which trigger is used?

A. Event trigger
B. Schedule trigger
C. Manual trigger
D. Tumbling trigger

Q18.

Data should load when a file arrives in Blob storage. Which trigger is best?

A. Schedule
B. Event-based trigger
C. Manual
D. Debug

Q19.

Which Integration Runtime is used for on-premises data sources?

A. Azure IR
B. Self-hosted IR
C. SSIS IR
D. None

Q20.

You are testing a pipeline before deployment. Which feature is used?

A. Trigger run
B. Debug mode
C. Publish
D. Monitor

Section 6: Monitoring & Management

Q21.

A pipeline fails during execution. What is the first step?

A. Delete pipeline
B. Check logs and error details
C. Restart system
D. Ignore

Q22.

Which feature helps track pipeline execution history?

A. Author tab
B. Monitor tab
C. Dataset tab
D. Trigger tab

Q23.

To get notified when a pipeline fails, what should be configured?

A. Logging only
B. Alerts and notifications
C. Manual checks
D. No action

Q24.

A pipeline is running slowly due to large data volume. What is the best optimization?

A. Reduce data manually
B. Parallel processing and partitioning
C. Stop pipeline
D. Ignore

Q25.

Which approach helps detect performance bottlenecks in pipelines?

A. Ignore metrics
B. Use monitoring metrics and logs
C. Delete pipeline
D. Manual guess